

The empirical recurrence for OEIS A232946, recovered from its tail

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Abstract

OEIS A232946 records “Empirical recurrence of order 57” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 122$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 60$. Its last coefficient is $c_{57} = 1 \neq 0$, so no further tail satisfies anything shorter; any order-57 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A232946 is “Number of $n \times 6$ 0..3 arrays with no element $x(i,j)$ adjacent to value $3-x(i,j)$ horizontally, vertically or antidiagonally, and top left element zero.”. It has offset 1 and begins

243, 19601, 1905737, 190528543, 19245039957, 1950478851269, 197901829874065, 20086980767566967, ...

The entry states:

Empirical recurrence of order 57 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 07:45:10, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 122$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 122$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 57: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 57.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 244$ exact terms from $n = 3$ on, it returns order 57 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{57} c_i a(n-i),$$

c_1	160	c_{16}	−13034500604990	c_{31}	−1279450694415924	c_{46}	−59901293102
c_2	−7026	c_{17}	−89317709774098	c_{32}	−16342780540878970	c_{47}	22781056479
c_3	118279	c_{18}	82014208755454	c_{33}	7506304064867354	c_{48}	−6040948419
c_4	−794979	c_{19}	437016054434155	c_{34}	4152459460419021	c_{49}	1282626046
c_5	812127	c_{20}	−323144206016938	c_{35}	−4693119694328971	c_{50}	−218928881
c_6	22072781	c_{21}	−1755574504370428	c_{36}	1001466819368157	c_{51}	30667342
c_7	−197127985	c_{22}	1168920154024324	c_{37}	689205080421573	c_{52}	−3503663
c_8	703929182	c_{23}	5378992122558144	c_{38}	−592609227982099	c_{53}	317400
c_9	1466889338	c_{24}	−4132984951938283	c_{39}	213432238065702	c_{54}	−22795
c_{10}	−24446478523	c_{25}	−11484650126276149	c_{40}	−34851273689710	c_{55}	1310
c_{11}	73816164148	c_{26}	11486201292827614	c_{41}	−8292076222369	c_{56}	−53
c_{12}	182528525668	c_{27}	15449511686856928	c_{42}	7688012759378	c_{57}	1
c_{13}	−1548888392982	c_{28}	−21494466442505010	c_{43}	−2251641061848		
c_{14}	438052060432	c_{29}	−10366787096877708	c_{44}	291111457927		
c_{15}	14493979822045	c_{30}	25026277126535480	c_{45}	59108991667		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 60$, and it is the recurrence OEIS A232946 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{57} - c_1 x^{56} - \dots - c_{57}$. Its constant term is $-c_{57} = -1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 57 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 57, so $\deg q = 57$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 57, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 12 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 57, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-57 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A232946.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.